

Experimental Design

CML Week 8

2026-10-27

Schedule

1. Design-based causal inference & Monte Carlo simulation
2. Covariate adjustment
3. Balancing weights
4. Double robustness, Double ML and TMLE
5. Heterogeneous Treatment Effects
6. HTE II: Neural Networks
7. Off-policy evaluation and optimization
8. **Experimental design**
9. Panel data and the new diff-in-diff
10. Partial Identification
11. Response adaptation
12. Interference



From analysis to design

Thus far: How to *analyze* data to estimate causal effects

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This week: How to *design* experiments to make causal inference easier

> To consult the statistician after an experiment is finished is often merely to ask him to conduct a post mortem examination. He can perhaps say what the experiment died of.
> — Ronald Fisher



Why design matters for experiments

Even in randomized experiments:

- Finite samples can have covariate imbalance
- Imbalance reduces precision
- Imbalance can bias subgroup analyses

Key insight: Design can achieve what analysis cannot

- Better balance $\Longrightarrow$ more precise estimates
- Careful randomization $\Longrightarrow$ valid inference
- Ex-ante optimization $\Longrightarrow$ avoid post-hoc problems

What is balance?

Covariate balance: Similarity between treatment and control groups on observed covariates

A formal definition: For treatment assignment $A_i \in \{0, 1\}$ and covariates $X_i \in \mathbb{R}^d$:

$$\text{Imbalance} = \left\| \frac{1}{n_1} \sum_{i:A_i=1} X_i - \frac{1}{n_0} \sum_{i:A_i=0} X_i \right\|$$

Why it matters:

- Balanced experiments have lower variance in treatment effect estimates
- Balance on prognostic covariates is especially valuable
- Balance facilitates valid subgroup analyses

How does it work?

The model: Suppose outcomes follow:

$$Y_i = \tau A_i + \beta X_i + \epsilon_i$$

where X_i is a prognostic covariate and ϵ_i is noise with $Var(\epsilon_i) = \sigma_\epsilon^2$.
Variance without adjustment:

$$Var(\hat{\tau}_{DM}) = Var(\bar{Y}_1 - \bar{Y}_0) = \left(\sigma_\epsilon^2 + \beta^2 Var(X)\right)\left(\frac{1}{n_1} + \frac{1}{n_0}\right)$$

Variance under perfect balance:

$$Var(\hat{\tau}_{balanced}) \approx \sigma_\epsilon^2\left(\frac{1}{n_1} + \frac{1}{n_0}\right)$$

Adjustment removes variation due to X, leaving only ϵ.
Balance gives adjustment for free

Balance in randomization

Complete randomization: Flip a coin for each unit

- Balance in *expectation*: $\mathbb{E}[X \mid A = 1] = \mathbb{E}[X \mid A = 0]$
- But not in any particular realization!

The problem:

- With 1 covariate, imbalance is $\mathcal{O}(1/\sqrt{n})$
- With d covariates, worst-case imbalance is $\mathcal{O}(\sqrt{\log d}/\sqrt{d})$
- Imbalance can be substantial in practice

The design challenge

Two extremes:

1. **Deterministic assignment:** Perfect balance, but no randomization
   - Can't do valid inference
   - Vulnerable to confounding
2. **Pure randomization:** Valid inference, but poor balance
   - High variance
   - Unlucky realizations possible

The goal: Achieve better balance while preserving randomization

- Restricted randomization
- Optimization-based designs
- Sequential adaptive designs

(Greevy et al. 2004)

Foundation: Greevy et al (2004)

The key insight: Match units into pairs before randomizing within matched sets
The approach:

1. Use distance-based matching on covariates
2. Form matched pairs with similar units
3. Randomize treatment assignment within each pair

Benefits:

- Combines matching (good balance) with randomization (valid inference)
- Reduces variance by ensuring prognostic similarity
- Permits straightforward paired analysis

The Greevy approach

Step 1: Define distance

- Multivariate distance on covariates
- Typically Mahalanobis distance: $d(i, j) = (X_i - X_j)' \Sigma^{-1} (X_i - X_j)$
- Can incorporate different weights for different covariates

Step 2: Optimize matching

- Find pairs that minimize total distance
- Subject to constraint: equal numbers in treatment and control

Step 3: Randomize

- Within each matched pair, flip a coin
- One unit to treatment, one to control

Greevy's optimization problem

Objective: Minimize total distance in matched pairs

$$\min \sum_{p=1}^{n/2} d(i_p, j_p)$$

subject to:

- Each unit appears in exactly one pair
- Total treatment assignments = $n/2$

Connection to combinatorial optimization:

- Minimum weight perfect matching on the complete graph of n units (*non-bipartite*: anyone can pair with anyone)
- Solved exactly by Edmonds' blossom algorithm in $\mathcal{O}(n^3)$ time, plus $\mathcal{O}(n^2)$ memory for the distances
- Scales to moderate-sized experiments

(Efron 1971)

Accidental bias

Efron's worry: a covariate u we didn't measure (e.g. a time trend)
Setup: $Y_i = \tau A_i + u_i$, with $n/2$ units in each arm

$$\begin{aligned}\hat{\tau}_{DM} - \tau &= \bar{Y}_1 - \bar{Y}_0 - \tau = \bar{u}_1 - \bar{u}_0 \\ &= \frac{1}{n/2} \sum_{i:\, A_i=1} u_i \;-\; \frac{1}{n/2} \sum_{i:\, A_i=0} u_i \\ &= \frac{2}{n} \sum_i z_i u_i = \frac{2}{n}\, z'u, \qquad z_i = 2A_i - 1 \in \{-1, +1\}\end{aligned}$$

- The treatment effect cancels: all that's left is the difference in u between the arms
- z_i is just the sign: $+1$ adds u_i with the treated, -1 subtracts it with the controls

From bias to variance

$$\hat{\tau}_{\mathrm{DM}} - \tau = \frac{2}{n} z'u$$

For the observed assignment, this is a **bias**. Averaged over draws, it is **variance**:

$$\begin{aligned}\mathbb{E}\left[(\hat{\tau}_{\mathrm{DM}} - \tau)^2\right] &= \frac{4}{n^2}\,\mathbb{E}\left[(z'u)'(z'u)\right] = \frac{4}{n^2}\,\mathbb{E}\left[u'zz'u\right] \\ &= \frac{4}{n^2}\,u'\,\mathbb{E}[zz']\,u = \frac{4}{n^2}\,u'\,\mathrm{Cov}(z)\,u\end{aligned}$$

- u is fixed, so it comes out of the expectation: only the assignment z is random
- $\mathbb{E}[z] = 0$ (each arm is equally likely), so $\mathbb{E}[zz'] = \mathrm{Cov}(z)$

Accidental bias: the worst case over covariates nobody anticipated

$$\max_{\|u\|=1} u'\,\mathrm{Cov}(z)\,u = \|\mathrm{Cov}(z)\|_2 = \lambda_{\max}\Big(\mathrm{Cov}(z)\Big)$$

- This max is the definition of the spectral norm; for a covariance matrix it is the largest eigenvalue, attained at the principal eigenvector

Coin flips are minimax

Every design pays the same total: $\mathrm{Var}(z_i) = 1$ for every unit, so

$$\operatorname{tr} \mathrm{Cov}(z) = n$$

› The eigenvalues of $\mathrm{Cov}(z)$ average 1, so $\lambda_{max} \geq 1$ for *any* design

Independent coin flips: $\mathrm{Cov}(z) = I$, so every direction gets exactly 1

› No design does better against a covariate you didn't anticipate

› Efron's conclusion: pure randomization *minimizes* the worst-case accidental bias

Pushing $\mathrm{Cov}(z)$ down in the directions of X pushes it up somewhere else

Example: matched pairs

Greevy's design: within each pair (i, j), $z_j = -z_i$

$$\mathbb{E}\left[(z'u)^2\right] = \sum_{\text{pairs } (i,j)} (u_i - u_j)^2$$

compared with $\sum_i u_i^2$ under independent coin flips

- u **similar within pairs** (what matching aims for): eigenvalue 0, the variance vanishes
- u **differs within pairs** ($u_i = -u_j$): eigenvalue 2, the variance *doubles*

Accidental bias of matched pairs: $\lambda_{max} = 2$

- That is the price of balance: a bet that the covariates you matched on are the ones that matter

(Harshaw et al. 2019)

The robustness–balance tradeoff

Exact design-based identity: Horvitz–Thompson with $\Pr[A_i = 1] = 1/2$ gives

$$\hat{\tau}_{HT} - \tau = \frac{2}{n}\, z'\mu, \qquad \mu_i = \frac{y_i(1) + y_i(0)}{2}$$

$$\mathrm{Var}(\hat{\tau}_{HT}) = \frac{4}{n^2}\, \mu'\, \mathrm{Cov}(z)\, \mu$$

This is Efron's formula with μ in the role of the nuisance. Split $\mu = X\beta + \varepsilon$ for covariates $X \in \mathbb{R}^{n \times d}$:

- **Balance:** make $\mathrm{Cov}(z)$ small along the columns of X, so $z'X \approx 0$
- **Robustness:** keep $\lambda_{max}\big(\mathrm{Cov}(z)\big)$ small, to protect against ε
- $\mathrm{tr}\, \mathrm{Cov}(z) = n$ means we cannot have both at once

The Gram–Schmidt Walk design

One parameter $\phi \in (0, 1]$ chooses the point on the tradeoff. With $\xi = \max_i \|x_i\|$:

$$\mathrm{Cov}(z) \preceq \left(\phi I + \frac{1-\phi}{\xi^2} XX'\right)^{-1}$$

- **Robustness:** $\lambda_{max}\Big(\mathrm{Cov}(z)\Big) \leq 1/\phi$
- **Balance:** directions in the span of X are shrunk by the XX' term

Variance bound:

$$\mathrm{Var}(\hat{\tau}_{HT}) \leq \frac{4}{n^2} \min_{\beta} \left[\frac{1}{\phi}\|\mu - X\beta\|^2 + \frac{\xi^2}{1-\phi}\|\beta\|^2\right]$$

- A ridge regression of μ on X, done by the design rather than the analysis
- $\phi = 1$: coin flips, Efron's minimax design. $\phi \to 0$: maximal balance, accidental bias up to $1/\phi$

This week's trajectory

The evolution of balance in experimental design:

1. **(Greevy et al. 2004):** Matching before randomization
   - Pairs of similar units, randomize within pairs
2. **(Morgan and Rubin 2012):** Rerandomization
   - Check balance, reject and re-randomize if poor
3. **(Kallus 2018):** Optimization-based balance
   - Directly optimize over randomization distributions
4. **(Harshaw et al. 2019):** Optimal linear balance
   - Balancing with strong theoretical guarantees and formal robustness tradeoff

References

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